

Harvesting the Sky:
Wing-Integrated Wind Turbines for Energy Recovery
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Introduction

As the aviation industry moves toward electrification, one of its greatest challenges continues to be the limited range of its low energy density of the current batteries. Martin Hepperle, Group Research Lead of the German Aerospace Center, explains, “In order to power larger aircraft a dramatic improvement in battery technology would be required,” energy density must increase by tenfold to attract commercial interest (Hepperle, 2012). This technological limitation underscores a critical barrier to making electric flight competitive with conventional aviation; intensifying the search for alternative energy solutions capable of extending flight range. Simultaneously, the environmental cost of aviation is becoming increasingly evident. Karie Riley and colleagues from the U.S. Environmental Protection Agency found that “UFP [ultrafine particulate pollution] concentrations are elevated downwind of commercial airports, and... proximity to an airport also increases particle number concentrations within residences” (Riley et al., 2021). This demonstrates how traditional turbine-engine aircraft significantly contribute to UFP, which is linked to respiratory inflammation and lung cancer. The dual challenges of energy inefficiency and environmental degradation establish a pressing need for sustainable transformation within aviation.

To combat these issues, I propose Wing-Integrated Wind Turbines (WIWTs). These are mechanical energy-harvesting devices embedded directly into the structural profile of an aircraft's wing. It functions by capturing the high-velocity airflow moving over the wing's surface and using that kinetic energy to spin turbine blades connected to an internal generator. This system is being introduced to serve as an onboard power supplement, designed to recharge propulsion batteries during flight to mitigate the current limitations of battery energy density.

Literature Review

Due to the persistent challenge of low battery density, researchers are now exploring onboard energy-harvesting systems to supplement stored power and extend the limited range of electric aircraft. Dr. Magedi Saad, an academic researcher at Tun Hussein Onn University of Malaysia, and colleagues note that “the Ram Air Turbine (RAT) is used as an emergency power source to feed vital apparatuses required for aircraft in case of main systems failure,” emphasizing its value as a “wind harvesting technology for aircraft” (Saad et al., 2021). Their observation demonstrates that wind-based energy generation is already a functional concept in aviation, even when confined to short-term emergency operations. Expanding on this, Matthew Dowling and Mark Costello from the Georgia Institute of Technology observe that “only large-scale systems typically achieve a power coefficient of over 50%. At smaller scales, power coefficients usually drop dramatically due to... low Reynolds numbers” (Dowling & Costello, 2018). The Reynolds number measures how air behaves around a blade, indicating whether the airflow stays attached or becomes weak and prone to separation. Showing the decline mirrors the aerodynamic constraints limiting RAT performance, revealing a shared efficiency challenge rooted in scale rather than principle. Adding a practical dimension, Dale Walter-Robinson, CEO of ElectronAir LLC, patented an *Energy cell Regenerative System for Electrically Powered Aircraft*, in which, “at least one fan generator [is] mounted on an aircraft... near a propeller... to harness the kinetic energy of the propeller blast,” so it doesn’t affect the flight characteristics of an aircraft (Walter-Robinson, 2014). This system demonstrates how continuous, drag-neutral energy harvesting can be structurally integrated into existing aircraft designs. Together, these studies show that the main challenge to effective airborne wind harvesting lies in engineering efficiency rather than conceptual feasibility. Overcoming small-scale aerodynamic losses through

integrated turbine design could transform wind systems from emergency backups into energy sources that extend the range of electric aircrafts.

Building upon RATs, recent research has shifted toward refining aerodynamic geometry and flow interaction to enhance overall power output. Dr. Magedi Saad and his research team conducted a Computational Fluid Dynamics (CFD) investigation that found “the counter-rotating [RAT] improved the performance of the conventional... RAT by increasing efficiency up to 45%, with power outputs reaching 113.26 kW compared to 62.36 kW” (Saad et al., 2021). While this study centers on RATs, the findings highlight how flow symmetry and wake control, key aspects of aerodynamic efficiency, could be applied in distributed turbine systems across aircraft to reduce drag and improve overall energy capture. Further supporting this potential, Aeronautical Engineering researcher, Miguel Bolaños-Vera and colleagues developed a CFD-based model of a commercial RAT featuring “a 1.016 m diameter rotor and three aerodynamic blade sections... [achieving] power generation within a 5-60 kW range by combining high-lift and low-drag airfoils” (Bolaños-Vera et al., 2024). Their segmented blade design, which tailors the geometry to local airflow conditions, mirrors the principles required for distributed wind systems integrated along wing structures, where dynamic wind-circulation regions demand adaptive airfoil design. Further, Francesco Trevisi, European Academy of Wind Energy PhD Award Winner, and colleagues explored the aerodynamic co-design of wings and turbines, reporting that “placing the turbines at the wing tips and rotating them inboard down lead to higher power production... [with] a maximum power coefficient $CP = 0.95$ at $\alpha = 9^\circ$ ” (Trevisi et al., 2025). Their work underscores how turbine-wing interaction can enhance, rather than hinder, aerodynamic efficiency, suggesting that energy-harvesting wings could become structural assets rather than performance liabilities. Collectively, these studies shift the discussion from

stand-alone turbine efficiency to aerodynamic synergy, showing that WIWTs can transform aircraft into hybrid energy-generating systems that sustain longer, more efficient flights.

As ideas advance toward integration into aircraft structures, optimizing the geometry, materials, and control systems of small-scale turbines is critical for maintaining efficiency and stability under variable conditions. Chris Kaminsky and fellow professors, from the Department of Engineering at Grand Valley State University, revealed that “increasing the attack angle of the airfoil created larger regions of separation causing what is known as the stall effect... [being] very inefficient because of the large amounts of drag that would exist” (Kaminsky et al., 2012). This study demonstrates that excessive flow separation at higher attack angles drastically reduces aerodynamic efficiency by increasing turbulence and drag. In the context of WIWTs, such inefficiencies could offset energy gains if not properly managed, emphasizing the need for designs that balance lift generation with flow stability. Shifting from aerodynamic performance to structural composition, Antonio Rosato and his team from the Department of Architecture and Industrial Design at the University of Campania Luigi Vanvitelli, report that “fiberglass is the most used material for blade manufacturing... [while] aluminum dominates vertical-axis models” (Rosato et al., 2024). As demonstrated by the quote, these materials are favored in small-scale designs due to their ability to minimize excess loss and create consistent energy generation. Extending beyond design and materials, Barzegar-Kalashani et al. from Swinburne University of Technology emphasized that “in wind turbines, the aerodynamic power can be increased or decreased by pitching the airfoil distributed across the blade” (Barzegar-Kalashani et al., 2023). This adaptive mechanism allows turbines to automatically adjust to fluctuating wind conditions, ensuring efficient operation and protecting the structure from overload. These studies reveal that successful WIWT implementation depends on the convergence of aerodynamic geometry,

lightweight materials, and intelligent control, which collectively enable steady power generation under the dynamic conditions of flight.

As WIWTs become feasible, establishing reliable electrical management and energy distribution systems is essential to utilize the harvested power. Shuangqi Li and colleagues, from the Department of Electronic and Electrical Engineering at the University of Bath, explain that by adding a small battery that uses just 5% of its capacity, more than 23% of the times the fuel cell was producing over 40 kW (high power) can instead be shifted to a more efficient range of 10-30 kW, helping the system run at its best performance (Li et al., 2021). This demonstrates how intelligent energy management can stabilize fluctuating power inputs, a principle directly applicable to storing and distributing variable wind energy from WIWTs. Sumantra Bhattacharya, Faculty of Engineering at Universität Ulm, and professors note that “the converter is divided into two separate stages...the resonant converter stage and the voltage-balancing stage,” the resonant converter stage can transfer electricity both ways between the high- and low-voltage systems, while the voltage-balancing stage adjusts the voltage to reach the proper level for the aircraft’s main power network (Bhattacharya et al., 2022). This two-stage converter framework enables multi-directional energy flow between propulsion, storage, and auxiliary systems, ensuring power is delivered efficiently and safely throughout the aircraft. Luís Praça, a researcher at the Federal University of Ceara, and colleagues state that “[smart power electronics converters] make it possible to keep the on-board microgrid stable and reliable as more energy resources and electronic loads are connected” (Praça et al., 2023). Praça emphasizes that advanced converters and energy management strategies are critical for maintaining stability and reliability in aircraft microgrids, ensuring that variable energy output can be effectively integrated without disrupting the system. Together, these sources illustrate that successful WIWT

integration relies on the convergence of adaptive energy storage, multi-directional conversion, and advanced microgrid control, ensuring wind-generated energy is efficiently stored, conditioned, and distributed across aircraft electrical systems.

Recent advances in aerodynamic modeling have shown that careful integration of propulsion and energy-harvesting systems into wing structures can transform aerodynamic interference into performance gains. Raymond Akagi, from California Polytechnic State University, shows that to reduce the effects of swirl and maximize efficiency, the actuator disk's diameter and rotation speed should be set to achieve a tip speed ratio just below 5 for maximum power (Akagi, 2021). This relationship between swirl control and power efficiency shows that WIWTs must manage wake behavior to minimize drag while maintaining smooth, efficient airflow across the wing. Building on this, Matthew Clarke, Aerospace Professor at Stanford University, and fellow professors, identify that “on the wing section outboard of the propeller... the local wing section experiences an increase in the effective angle of attack due to the induced upwash from the blades,” and the opposite is seen in the wing section inboard from the propeller (Clarke et al., 2021). These findings show that airflow around rotating systems can boost or reduce lift depending on placement, suggesting turbines in upwash zones could enhance wing performance and energy generation. Complementing this, Karen Deere, from NASA Langley Research Center, and colleagues, found that at a small angle of attack (0.6°), the cruise propulsors at the wingtip lower the wing's total drag by 7.5%, with the drag coefficient dropping to 0.0301 when powered (Deere et al., 2017). This measurable drag reduction demonstrates that aerodynamic coupling between propulsion and lift systems can be leveraged to achieve mutual benefit. Collectively, these studies illustrate how design precision - through controlled swirl, optimized placement, and wingtip interaction - lays the groundwork for successful WIWT

integration, transforming aerodynamic challenges into energy-producing opportunities for next-generation electric aircraft.

Across the reviewed studies, the literature collectively demonstrates that advancements in sustainable aviation hinge on balancing aerodynamic performance, structural integrity, and energy regeneration. Research on small-scale wind systems, turbine geometry, materials, and control technologies has shown that energy capture can be efficiently achieved under variable aerodynamic conditions. Meanwhile, studies on distributed propulsion and aerodynamic integration emphasize that embedding energy systems within lifting surfaces can reduce drag and enhance stability. These findings establish a foundation for combining aerodynamic efficiency with renewable energy recovery. Yet a critical research gap persists: WIWTs have never been directly examined in the academic literature, and no existing study investigates how a turbine system embedded within a lifting surface could influence both energy regeneration and aerodynamic performance. This study addresses that gap by asking: To what extent will WIWTs affect the flight range of electric aviation? It is hypothesized that properly designed WIWTs, in the Open Air Trailing Edge configuration, will enhance the operational range of electric aircraft by continuously supplementing onboard power with minimal aerodynamic penalty.

Methods

The study used computational fluid dynamics (CFD) to investigate how WIWTs influence the aerodynamics of electric aircraft. The analysis focused on understanding the effects of turbine placement along the wing and the role of different airfoil geometries on both aerodynamic performance and energy extraction. By simulating airflow, pressure distribution, and turbine power output, I evaluated how design variations affected lift, drag, and potential electrical power recovery. This CFD-based approach allowed controlled testing of multiple configurations, enabling direct comparisons between turbine locations and airfoil types. The resulting data provided a basis for correlating aerodynamic outcomes with turbine positioning and wing design, which is critical for optimizing WIWT integration.

This study began by constructing detailed 3D CAD models, following the structural and geometric modeling recommendations from Dr. Magedi Saad and colleagues, who evaluated six airfoils of varying aerodynamic qualities to study turbine interactions (Saad et al. 2021). Based on their framework, three airfoils - NACA 4412, CLARK Y, and a NACA 23015 - were selected to represent a range of camber and thickness distributions relevant to WIWT integration. Consistent with work from Esteben Valencia and fellow professors at the National Polytechnic Institute, the CAD workflow used parametric modeling to allow rapid reconfiguration of chord length, turbine inlet geometry, and internal cavity structure (Valencia et al. 2020). In continuation, the turbine was placed at three different positions per airfoil - leading-edge, mid-chord, and trailing edge - this decision was informed by Samuel Merryisha and Parvathy Rajendran, from the Engineering Campus at Universiti Sains Malaysia, whose advanced CFD research on lifting surfaces demonstrated how pressure gradients and surface shear vary predictably along a wing (Merryisha & Rajendran 2019). Furthermore, another test was

conducted where the turbine system was integrated into an internal tunnel cutout inside the wing. Two of these configurations were evaluated: one in which airflow passed through a forward section of the wing, and another in which airflow was routed through the full wing section across the top and then also across the bottom. For all three tunnel configurations, simulations were conducted solely with single turbine cases, where the turbine was placed in the middle of the cutout. Lastly, the turbine rotor itself was represented using a simplified geometry consistent with actuator-disk-based CFD recommendations from Eric Lynch, School of Aerospace Engineering at Georgia Institute of Technology, who emphasized that reduced-order rotor representation maintains realistic momentum extraction while keeping computational cost manageable (Lynch, 2011). All final wing-turbine assemblies were exported as STEP files for meshing.

Computational meshes were generated using a hybrid strategy reflecting the meshing principles outlined by Saad et al., Valencia et al., and Lynch, all of whom stress the necessity of resolving high-shear regions, boundary layers, and wakes to ensure numerical accuracy. The external fluid domain extended ten chord lengths upstream, fifteen downstream, and ten chord lengths above and below. Unstructured tetrahedral cells were applied in the far field, while structured prism inflation layers were used near walls and the turbine cavity. Eric Lynch emphasized the importance of accurately capturing surface shear and boundary-layer growth in turbine-airfoil interaction studies, guiding the use of inflation layers targeting $y^+ \approx 1$ (Lynch, 2011). Local refinement was applied near blade tips and at the turbine inlet region, to resolve tip vortices and steep velocity gradients. Mesh skewness was kept below 0.7, and grid independence was verified using less than a 5% variation between medium and fine grids of the NACA 4412 mid-chord (T-M) position (Lynch, 2011; Saad et al. 2021).

To accurately simulate airflow around the wing and through the embedded turbine, the SST k- ω turbulence model was implemented consistent with the methodology of Chris Kaminsky and companions. Their work demonstrated that the SST k- ω model provides superior performance in capturing boundary-layer behavior, flow separation, tip vortices, and transitional structures over a wide range of aerodynamic regimes (Kaminsky et al. 2012). Solver settings used dynamic mesh and User Defined Functions (UDFs) to represent rotor motion and physical properties (Saad et al. 2021). Similarly, this study employed a dynamic mesh to simulate turbine rotation, with the UDF specifying rotor mass, rotational axis, and blade motion. The solver was configured as a pressure-based, three-dimensional, transient simulation using absolute velocity formulation. Residuals, lift and drag forces, and torque were monitored continuously to verify steady-state convergence (Kaminsky et al. 2012). The combined turbulence-solver-actuator disk framework ensured that complex aerodynamic behaviors were fully resolved across all geometric configurations.

Each airfoil-turbine configuration was simulated under controlled boundary conditions matching the domain-sizing and inflow recommendations of Esteban Valencia and fellow professors. They emphasized that consistent boundary conditions and sufficiently large flow domains are essential for avoiding numerical interference and enabling realistic flow development (Valencia et al. 2020). Thus, a uniform inlet velocity of 175m/s was applied at the upstream boundary, a pressure-outlet condition was enforced downstream, and all wing and turbine surfaces were assigned no-slip wall conditions. These settings prevented artificial acceleration or recirculation near the boundaries and ensured meaningful flow distribution. Domain sizing was maintained based on Valencia et al. to prevent the outer flow from constraining wake growth; while the dynamic mesh and UDF-driven rotor motion followed Saad

et al.'s method for representing rotating machinery in a transient, pressure-based solver environment (Saad et al. 2021; Valencia et al. 2020). Each configuration ran until aerodynamic forces and rotor torque indicated steady-state behavior.

After simulation completion, data was post-processed to determine the aerodynamic effects of turbine placement and airfoil geometry on wing performance and energy extraction. Lift, drag, local velocity, wattage, and turbine torque were extracted and compared across all of these configurations; further, percent power efficiency according to Betz's Law, percent decrease in lift, and percent increase in drag were calculated using these values. Turbine-induced wake structures and inflow velocity deficits were examined using actuator-disk theory and momentum methods (Lynch, 2011). Finally, velocity-field comparisons and surface-pressure differences across airfoils incorporated principles regarding surface modification effects and interpretation of WIWT-induced aerodynamic modifications (Merryisha & Rajendran, 2019).

The methodologies described above provided a clear framework for investigating the effects of WIWTs on electric aircraft aerodynamics. The outlined approaches from the sources shaped this study, allowing for a systematic and controlled evaluation of turbine placement and airfoil geometry. The resulting datasets will continue to support analysis of flight range and turbine efficiency, demonstrating how careful application of research can make a complex engineering problem manageable.

Data Analysis

It was hypothesized that properly designed turbines in the Open Air Trailing Edge (T-B) configuration would enhance the operational range of electric aircraft by continuously supplementing onboard power with minimal aerodynamic penalty. However, the resulting data required a definitive denial of this hypothesis, because the T-B location triggered a catastrophic aerodynamic failure across all airfoil models. For example, the NACA 4412 in the T-B position generated a significant power output of 2,840 Watts, representing 56.5% efficiency, but this energy gain was completely negated by a 536.4% increase in drag and a 34.9% decrease in lift, resulting in a net energy loss for the aircraft. Overall, the T-B placement specifically proved that the power-to-drag ratio is fundamentally unsustainable for long-range flight, especially when configurations like the leading edge (T-F) produced turbine efficiency in the low 60s rather than the mid 50s.

Airfoil thickness serves as the exclusive catalyst for power generation, while variations in camber fail to provide any significant "aerodynamic buffer" for the wing's lifting capacity. The physical requirement for air to traverse a longer path over a curved surface creates a localized high-velocity field, and the data suggests that the magnitude of this acceleration is determined solely by the thickness of the profile rather than by the degree of its curvature. For instance, the NACA 23015, the thickest profile in the study, achieved a peak power output of 7,881 Watts at the T-F position. When compared to the thinner Clark Y airfoil, which produced 6,023 Watts at the same location, the thicker profile shows a 30.8% power advantage. This performance gap is even more pronounced than the baseline ideal power of 2,647.6 Watts, proving that geometric width is the fundamental prerequisite for energy harvesting. This result bridges the gap in the research question by identifying "thickness" as the only metric that effectively uses the wing's

curve to achieve higher velocities. Furthermore, the data regarding lift loss at the T-B position proves that camber does not grant the wing any special resilience against integration penalties. Many aerodynamic theories suggest that a higher camber might "stiffen" the pressure field, but the simulations show a near-identical collapse across different shapes. Specifically, the low-camber Clark Y at the T-B position dropped the baseline lift coefficient (CL) from 1.02 to 0.663, a 35% decrease in lift. Similarly, the highly cambered NACA 4412 dropped from 1.08 to 0.713, representing a 33.98% decrease. The fact that a flat-bottomed wing and a high-lift wing have only a minor 1% difference of their lift loss at the rear position demonstrates that the degree of camber is irrelevant to the wing's "fragility." This suggests that the structural cutout for the turbine disrupts the boundary layer uniformly across all airfoils, such that the airfoil's specific curvature cannot prevent changes in aerodynamic values. Therefore, for auxiliary power systems to be viable, designers must focus on maximizing thickness for power while accepting that camber won't save the aircraft from the lift-loss penalties.

Internal tunnels designed to hide turbines within the wing structure create a high-pressure stagnation barrier that physically chokes the generator's ability to spin. While internal integration is often proposed to minimize external drag, the data proves that these configurations are largely ineffective for power generation. The internal front (S-F) configuration for the CLARK Y reached only 76,500 rotations per minute and generated a meager 381 Watts. By comparison, the open-air T-F equivalent position for the same airfoil reached 148,500 rotations per minute and 6,023 Watts. This instance isn't solely dedicated to the CLARK Y airfoil, as both the NACA 23015 and 4412 have identical representations, differing only in numerical values. This massive discrepancy addresses the gap in research concerning hidden turbines by proving that internal ducting creates a "pressure wall" that prevents air from entering the intake at the required

velocities. Additionally, the local velocity inside the NACA 23015 S-F tunnel was measured at only 136 m/s, and the T-F configuration had a local velocity of 248 m/s, further proving this idea. Continuously, pressure-gradient data confirms that a high-pressure air bubble sits at the entrance to the tunnels, limiting the total mass flow rate through the turbine blades. Because the air cannot exit the narrow internal tunnel quickly enough, it effectively blocks new air from entering and starves the generator of the kinetic energy required for significant output. This phenomenon is consistent across all airfoil types: the Clark Y S-F configuration produced only 381 Watts, while the NACA 4412 S-F produced 453 Watts. Even the most efficient internal configuration, the bottom tunnel on the NACA 23015, only achieved 2,161 Watts, which is still 18 percent lower than the baseline ideal power. In addition, the mechanical efficiency of these internal systems peaked at only 48 percent for the bottom tunnels, while open-air placements consistently exceeded 56 percent. Leading to the conclusion that for auxiliary power to be viable, the turbine must be exposed to open airflow to avoid the stagnation effects of internal configurations.

The aerodynamic stability of an airfoil modified by a turbine integration does not follow a linear decline but instead adheres to a "U-shaped" performance distribution across the chord. This distribution identifies the mid-chord (T-M) as a protective "aerodynamic well" that minimizes structural interference with the wing's most sensitive pressure zones, the leading-edge suction peak, and the trailing-edge separation point. When isolating the CL as the primary metric for aerodynamic health, a distinct pattern of volatility emerges at the wing's extremities across all three test profiles. At the T-F, where air velocity is highest, the turbine acts as a physical obstruction at the stagnation point. This "blunts" the suction peak, which is responsible for most of a wing's lift. For the high-camber NACA 4412, a lift drop from a baseline of 1.08 to 0.821 (a 24% loss) resulted, while the thick NACA 23015 saw a drop from 0.81 to 0.638 (a 21.2% loss).

This initial penalty proves that the T-F position is too "crowded" with pressure activity to sustain a mechanical cutout without significant aerodynamic degradation. However, as the turbine is moved to the T-M position, the lift values show a remarkable recovery, suggesting that the airflow has successfully transitioned past the suction peak and regained a degree of laminar stability. In this "aerodynamic well," the NACA 23015 recovered to a CL of 0.722, and the Clark Y reached its most stable lift value of 0.908. This stability is abruptly lost as the turbine moves toward the T-B, where the Clark Y plummeted to 0.663 (a 35% total loss) and the NACA 23015 fell to 0.527 (a 34.9% loss). This secondary collapse is caused by the turbine acting as a catalyst for total boundary-layer detachment in a region where the airflow is already low-energy. The drag ratio provides further empirical validation of this U-shaped trend: for the NACA 4412, the drag penalty began at 3.615 at the front, dropped to its lowest point of 3.153 at the T-M, and then rose to 6.846 at the T-B. This consistency across diverse profiles proves that the T-M is the only location where the wing's geometry can structurally tolerate a cutout. By utilizing this "sweet spot," designers can harvest energy while maintaining a lift-to-drag ratio (C_D/C_L) that is 56% higher than the disastrous trailing-edge configurations.

The operational viability of WIWTs is fundamentally governed by the mechanical load profile across the airfoil, which is characterized by an inverse relationship between rotational frequency and torsional resistance. In this technical context, RPM (Rotations Per Minute) measures the sheer velocity at which the turbine cycles through the fluid stream to capture energy. Conversely, torque represents the mechanical "heaviness" or the rotational force exerted on the turbine's central shaft. While RPM dictates the potential for high-frequency electrical generation, torque determines the structural tax levied on the bearings and generator; excessive torque, particularly when coupled with high velocity, accelerates material fatigue and introduces

harmonic instabilities. The data reveals a stark chord-wise gradient, in which mechanical stress dissipates as the turbine moves toward the T-B position. At the T-F of the NACA 23015, the system reaches its most violent state, peaking at 159,400 RPM with a substantial torque of 0.47 Nm. While this maximizes raw power, the high torsional load at such extreme frequencies subjects the internal gears to "mechanical chatter," microscopic, high-velocity impacts that degrade the system's integrity. Moving to the T-B illustrates the opposite extreme of the external flow: for the NACA 4412, RPM drops to 117,000 while torque falls to a manageable 0.23 Nm. This outcome represents a 44% reduction in torsional stress compared to the T-F, modeling a "low-impact" environment that maximizes the turbine's operational lifespan at the expense of energy density. Additionally, the internal tunnel configurations provide a critical baseline for low-stress integration. The S-F position across all models shows the lowest mechanical burden, with torque values as low as 0.05 Nm, indicating that internalizing the flow significantly dampens the physical "jerk" on the rotor. However, the corresponding drop in RPM, to approximately 76,500 in the CLARK Y, demonstrates a trade-off in conversion efficiency. By analyzing this spectrum, it becomes clear that the feasibility of WIWTs is not merely a search for maximum power, but a precise calibration of the torque-to-speed ratio to ensure the electronics can process energy without suffering mechanical surges.

The ultimate physical feasibility of WIWT systems is governed by a fundamental "Energy Harvesting Paradox." This phenomenon reveals that the aerodynamic thrust required to overcome the added drag of a turbine often exceeds the electrical power harvested, resulting in a net energy deficit across all tested configurations. By analyzing the Thrust Power required for steady flight, the data exposes a significant "propulsion tax." For the NACA 23015 at the T-B, the thrust power required to maintain level flight at 175 m/s skyrocketed to 603,666.98 W, a

substantial increase from the 61,718.83 W baseline. When compared to the turbine's actual harvest of only 3,252 W in that same position, the result is a "Thrust-to-Power" ratio of roughly 185:1. This highlights a thermodynamic barrier in the current model of static cruise integration; rather than extending range, the turbine acts as an "aerodynamic anchor." Even in one of the most optimized scenarios, the NACA 4412 at the T-M, the thrust power requirement of 189,167.53 W leaves a net deficit of over 184,000 Watts compared to the 4,210 W produced. However, this does not indicate a failure of the technology, but rather a necessary shift in its operational mission. The data suggests that these massive drag spikes could be leveraged as a functional benefit if the system is used during flight phases where drag is used to slow the aircraft, such as landing. By demonstrating that turbines can reach nearly 62% of the Betz limit, the research proves the mechanical hardware is highly effective, provided the aerodynamic "cost of entry" can be mitigated through smarter integration.

The synthesis of the simulation data indicates that while energy harvesting from wings is physically possible and mechanically efficient, it is currently constrained by critical aerodynamic limitations. These observations identified the NACA 23015 T-M as the optimal integration point. This configuration represents the "Best in Class" because it successfully sits in the Aerodynamic Well, minimizing lift loss to 10.00% while yielding the most stable power output of 5,028 Watts. Unlike the leading-edge positions, which "choke" the suction peak, or the trailing-edge positions, which trigger catastrophic flow separation, the T-M utilizes the wing's pressure plateau to harvest energy with the lowest relative penalty. Nevertheless, several broader limitations must be addressed before commercial implementation can be considered. First, this study utilized a fixed freestream velocity of 175 m/s; however, commercial aircraft operate across a wide flight envelope. At higher transonic speeds, air compressibility becomes a dominant factor, and the

presence of a WIWT could trigger premature shockwave formation, exacerbating drag spikes. Additionally, the simulations assumed a localized, isolated turbine environment, which does not account for the span-wise flow interference or the cumulative drag penalty on a full-scale aircraft with multiple units. There is also the challenge of maintenance and contamination; in real-world environments, the intake tunnels and turbine blades would be susceptible to bird strikes, icing, and debris, which would rapidly degrade the Betz efficiency recorded in these idealized digital conditions. Ultimately, this research proves that the barrier to "Free Energy" in aviation is not the turbine's ability to spin, but the wing's ability to stay efficient while carrying it. The data reveals that if we can "fix" the integration aerodynamically, specifically by reducing the C_D/C_L ratio, the system is ideal for energy recovery.

Conclusion

The results of this investigation demonstrate that the original hypothesis was not supported. It was predicted that WIWT systems near the trailing edge, would generate supplemental electrical power with minimal aerodynamic penalty. However, this assumption proved incorrect because the drag created by turbine integration consistently required more propulsion power to overcome than the turbine could generate in return.

Beyond the rejection of the hypothesis, several key findings emerged. First, geometric thickness acted as a velocity amplifier that increased turbine power output. Second, camber did not provide aerodynamic protection against integration penalties. Third, internal placements restricted airflow and reduced turbine effectiveness. Fourth, the mid-chord turbine integration was found to minimize inefficiencies compared to forward or rearward placements. Finally, mechanical feasibility depended not only on RPM but on maintaining a stable torque-to-speed ratio to avoid structural fatigue. However, across every configuration, the aerodynamic penalty exceeded the power generated by the turbines. Together, these findings confirm that while WIWT systems are mechanically capable, aerodynamic compatibility remains the central obstacle.

To address this challenge, one solution is the strategic use of regenerative energy capture during descent and landing. Aircraft generate drag in these phases to reduce speed and altitude. Integrating WIWT systems into this operational window would transform unavoidable drag into productive energy recovery. Instead of functioning as a propulsion burden, the turbines would operate when aerodynamic resistance is already necessary. The recovered electricity could power onboard systems or recharge batteries for hybrid aircraft. This solution benefits overall system

efficiency because it aligns turbine operation with natural flight dynamics rather than opposing them. A limitation, however, is that descent represents only a fraction of total flight time. The energy captured may not fully offset the added structural weight carried throughout climb and cruise, making lightweight materials and efficient storage essential for net benefit.

A second solution involves retractable WIWT architectures combined with adaptive flow management. By allowing turbines to deploy only during surplus-energy states and remain sealed during cruise, the aircraft can avoid continuous aerodynamic disruption. Supplementing this system with active flow control would smooth airflow around the aperture when deployed, reducing separation and drag spikes. This approach enhances overall viability because it preserves aerodynamic efficiency during steady flight while still enabling energy harvesting when conditions permit. The limitation lies in structural and mechanical complexity. Moving components, reinforcement structures, and control systems introduce new failure points and maintenance demands, potentially increasing certification difficulty and long-term operational cost.

Ultimately, this research shows that the barrier to effective airborne energy harvesting is not the turbine's ability to generate power, but the wing's ability to remain aerodynamically optimized. Future research should focus on variable-geometry airframes, lightweight composite integration strategies, and broader flight-envelope simulations to determine where energy recovery can complement aerodynamic performance. By identifying the NACA 23015 T-M configuration as a baseline, this study provides the mathematical foundation for a future where aircrafts do not just consume energy, but actively recover it.

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Appendix

This appendix presents the aerodynamic and power-related parameters measured for three airfoils, NACA 4412, Clark Y, and NACA 23015, across multiple turbine placement configurations. The lift coefficient (CL) quantifies the amount of lift generated by the airfoil, while the lift penalty (%) represents the percentage reduction in lift relative to the baseline (no turbine) condition. The drag coefficient (CD) measures the aerodynamic resistance experienced by the airfoil, and the drag penalty (%) indicates the increase in drag compared to the baseline. Thrust needed (W) refers to the power required to overcome aerodynamic drag and maintain steady motion. Local velocity (m/s) is the airflow speed at the turbine location, which directly influences energy capture. Maximum power (Betz, W) represents the theoretical upper limit of power extraction based on the Betz limit, while actual power (W) is the estimated real power produced by the turbine. Efficiency (Betz %) expresses the ratio of actual power to the theoretical maximum. Finally, RPM (revolutions per minute) and torque (Nm) describe the rotational speed and mechanical output of the turbine, respectively.

Airfoil	Configuration	Lift Coefficient (CL)	Lift Penalty	Drag Coefficient (CD)	Drag Penalty	Additional Thrust Needed (W)	Local Velocity (m/s)	Max Power (Betz, W)	Actual Power (W)	Efficiency (%) Betz	RPM	Torque (Nm)
NACA 4412	Baseline (No Turbine)	1.08	-	0.013	-	54705.32	-	-	-	-	-	-
	T-F (Leading Edge)	0.821	23.981	0.049	276.92	271245.74	236	10,958	6,480	59.13	151,700	0.41
	T-M (Mid-Chord)	0.961	11.019	0.04	207.69	189167.53	208	7,502	4,210	56.12	133,700	0.3
	T-B (Trailing Edge)	0.713	33.981	0.089	584.62	567296.98	182	5,030	2,840	56.46	117,000	0.23
	S-F (Internal Front)	1.015	6.019	0.029	123.08	129850.00	125	1,627	453	27.84	80,300	0.05
	S-T (Top Tunnel)	0.94	12.963	0.038	192.31	183723.94	156	3,161	1,303	41.22	100,300	0.12
	S-B (Bottom Tunnel)	0.896	17.037	0.044	238.46	223179.69	164	3,674	1,778	48.39	105,400	0.16

Airfoil	Configuration	Lift Coefficient (CL)	Lift Penalty	Drag Coefficient (CD)	Drag Penalty	Thrust Needed	Local Velocity (m/s)	Max Power (Betz, W)	Actual Power (W)	Efficiency (%) Betz	RPM	Torque (Nm)
CLARK Y	Baseline (No Turbine)	1.02	-	0.015	-	66834.56	-	-	-	-	-	-
	T-F (Leading Edge)	0.765	25.00	0.054	260.00	320805.88	231	10,278	6,023	58.60	148,500	0.39
	T-M (Mid-Chord)	0.908	10.98	0.046	206.67	230240.64	202	6,871	3,841	55.90	129,900	0.28
	T-B (Trailing Edge)	0.663	35.00	0.093	520.00	637498.87	178	4,710	2,652	56.31	114,400	0.22
	S-F (Internal Front)	0.949	6.96	0.033	120.00	158036.62	119	1,405	381	27.12	76,500	0.05
	S-T (Top Tunnel)	0.877	14.02	0.043	186.67	222832.67	152	2,925	1,199	40.99	97,700	0.12
	S-B (Bottom Tunnel)	0.826	19.02	0.049	226.67	269603.81	160	3,413	1,648	48.29	102,900	0.15

Airfoil	Configuration	Lift Coefficient (CL)	Lift Lost (%)	Drag Coefficient (CD)	Drag Gained (%)	Thrust Needed	Local Velocity (m/s)	Max Power (Betz, W)	Actual Power (W)	Efficiency (%) Betz	RPM	Torque (Nm)
NACA 23015	Baseline (No Turbine)	0.81	-	0.011	-	61718.83	-	-	-	-	-	-
	T-F (Leading Edge)	0.616	23.95	0.043	290.91	317247.16	248	12,711	7,881	62.00	159,400	0.47
	T-M (Mid-Chord)	0.729	10.00	0.035	218.18	218197.87	218	8,639	5,028	58.20	140,100	0.34
	T-B (Trailing Edge)	0.527	34.94	0.07	536.36	603666.98	191	5,807	3,252	56.00	122,800	0.25
	S-F (Internal Front)	0.77	4.94	0.025	127.27	147556.82	136	2,098	615	29.31	87,400	0.07
	S-T (Top Tunnel)	0.697	13.95	0.032	190.91	208654.23	167	3,878	1,598	41.21	107,400	0.14
	S-B (Bottom Tunnel)	0.664	18.02	0.037	236.36	253246.61	175	4,465	2,161	48.40	112,500	0.18